

Directional Collapse in Fine-Tuning: Preserving the Way Back

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Abstract

Paired tasks use the same examples in opposite directions, yet fine-tuning a language model on one direction can degrade its pre-existing ability in the other. We study this directional collapse through matched training comparisons across translation, code transformation, semantic parsing, factual relations, and formal tasks. The central question is whether reverse capability can be preserved by changing the direction of existing training pairs, rather than collecting additional data or introducing a specialized objective. We compare reversed-pair mixtures with forward-only training, generic replay, reverse-only training, and additional forward optimization. Recovery experiments distinguish behavioral loss from evidence of persistent capability. Extensions to cross-language code translation, Boolean reasoning, and unit conversion retain failed small-model gates. Larger-model unit-conversion replications improve both directions under forward-only training, providing a boundary case to collapse. The initial contrastive campaign is complete; registered scale and domain extensions continue. The methodological contribution is a testable comparison: evaluate both task directions, and compare proposed bidirectional objectives with ordinary supervised fine-tuning that accounts for directional information and computational cost. Quantitative conclusions remain pending integration and validation of the completed experiments.

1 Introduction

A paired dataset already contains two tasks. A translation pair can teach either language direction; a program transformation can be learned or undone; and a question–SQL pair can support semantic parsing or verbalization. Changing direction requires changing which side is the input, not collecting a second paired corpus. Nevertheless, training only one direction can damage the model’s ability to perform the other.

This phenomenon is established in particular settings. Nikiema et al. (2025) describe *cognitive specialization* in code obfuscation, and Zhu et al. (2024) examine direction-dependent degradation in translation. Our question is how broadly the failure occurs, what comparisons explain it, and how much machinery is needed to prevent it.

A failure that standard evaluation can miss. Improvement on the trained task does not establish that its inverse remains available. General benchmarks also need not measure this specific loss. We therefore evaluate forward and reverse success alongside general-ability probes, retaining null effects and model-specific boundary conditions. The completed results will quantify this comparison rather than assume that every model or domain collapses. We retain model–task–seed results separately and report the registered loss threshold descriptively before making aggregate inferential claims.

Direction as an experimental variable. Our main intervention replaces a share of forward pairs with their own reversal. Generic instruction replay, reverse-only training, and longer forward training test alternative explanations. These controls ask whether recovery depends on directional content, general rehearsal, reverse learnability, or additional optimization. Reversed pairs require no new paired examples, but budget and performance claims must be supported separately: replacement holds the number of instances fixed, while realized token counts, steps, and forward retention are audited rather than inferred from the word “replacement.”

Generality beyond the original domains. The current study covers natural-language, code, and formal transformations. Our next priority was to extend it to Python–C++ translation, Boolean formula–truth-table conversion, and unit conver-

sion. These tasks add executable semantics, logical equivalence, and numerical reasoning. Their exact scorers passed oracle audits, but all original small-model cells failed eligibility. Those failures provide boundary evidence about task specification and starting capability. Separately registered Gemma12B unit-conversion campaigns passed the unchanged gate and improved both directions under forward-only SFT across replication seeds. Separately registered explicit-output variants retain the original semantic splits and scorers and are exploratory diagnostics.

Does a specialized objective add value? Paired contrastive learning provides a direct test. Aligning representations may preserve reverse generation, but it also supplies both sides of each pair and uses additional compute. Our comparison therefore includes ordinary SFT with reversed pairs, additional-CE compute controls, and shuffled-pair ablations. A contrastive benefit would establish an objective-specific contribution; a null would establish the strength of a simpler baseline. Neither outcome is assumed in advance.

What does recovery tell us? Prompting, adapter scaling, relearning against a model-matched never-had control, and update repair provide complementary evidence about retained capability. We report them separately: behavioral recovery is evidence relevant to suppression, not proof that every observed loss is reversible. Information loss and inverse search difficulty are boundary conditions to investigate, rather than established universal limits on the remedy.

The practical test is simple: when fine-tuning on paired tasks, evaluate both directions; when attributing a gain to a bidirectional objective, compare against ordinary SFT while accounting for directional information and compute. This is a falsifiable comparison, not a presumption that auxiliary objectives are unnecessary. Our contribution is the joint preservation comparison: a cross-domain dose–performance frontier, exposure-accounted objective controls, and explicit boundary cases. The general-ability, independent-transfer, and robustness analyses in Appendix A are draft evaluation plans; their results are required before extending the corresponding claims.

2 Directional collapse, defined

A *paired task* is a pair of datasets (X, Y) and (Y, X) over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call $X \rightarrow Y$ the *forward* direction and $Y \rightarrow X$ the *reverse*. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately and never counted as success:

Echo the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

Off-target the output is in the wrong language, format, or notation.

Thresholds for continuous criteria are fixed from the *untouched* model’s own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

3 Setup

Domains. The audited main grid contains 11 task orientations covering translation, code transformation, semantic parsing, factual relations, executable reasoning, format conversion, and algebra. Opposite orientations of the same task are not independent domains. The invented-format never-had control is excluded from this count.

Models. The current audited results cover Llama 3B/8B, Gemma 4B/12B, and OLMo 1B. Eligibility and seed coverage vary by task and model; failed gates are retained.

Arms and budget. The forward-only baseline (SFT) trains on forward pairs. Each MIX k arm replaces $k\%$ of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count and optimizer steps are matched to SFT at every dose.

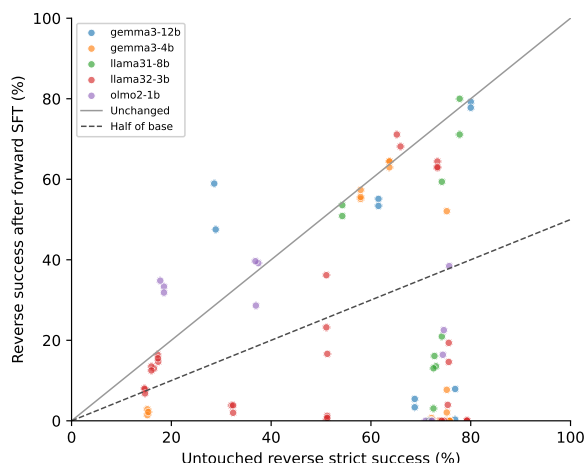


Figure 1: Each point is an audited model–task–seed cell. Both rates come from the same evaluation pass. The dashed line marks half of the base rate; repeated tasks and seeds are not independent observations.

We audit realized sequence and supervised-token counts separately: prompt length and unequal side lengths can change token exposure even when the pair count is fixed. Golovneva et al. (2024) distinguish data-matched from compute-matched reverse training; replacement requires no new paired examples. Any compute-saving claim is scoped to the measured training recipe and its audited token exposure.

Two controls separate directional loss from its alternatives. REPLAY replaces the same share with generic instruction data, so REPLAY — SFT is ordinary forgetting and MIX — REPLAY is what direction specifically buys. REV trains on reverse pairs only and is the ceiling every reverse number is read against; where REV is near zero the direction is not learnable in that cell and no null there is interpretable. The complete arm definitions and realized direction/token counts are retained in run manifests.

4 Measuring directional loss across tasks

The current snapshot contains 91 distinct audited model–task–seed cells from 103 validated evaluation campaigns. For repeated evaluations, we select the latest valid pass by completion time and retain earlier campaigns in the audit; selection does not use effect sign. In 52 cells, forward SFT leaves reverse strict success at most half the untouched model’s rate in that same pass. These counts apply a descriptive threshold; they are not pooled significance tests, independent-domain counts, or a universal failure claim.

Generic replay as a control. Every valid main pass includes generic replay where registered. Its contrast with reversed-pair SFT tests directional content beyond general rehearsal. Aggregate contrasts and their intervals remain to be integrated.

Instruction-following retention. General-ability probe artifacts are retained; their overlap and comparability audit remains pending. The draft analysis pairs changes in inverse success with changes in instruction following and general reasoning on a fixed panel (Appendix A.2). Probe instances must be checked against task training, replay, and filler data. No preservation or disproportionate-loss claim is made from task scores alone.

Moderators. Model family, size, task orientation, and starting capability are reported separately. Seed ranges will accompany partial-collapse claims.

5 Preserving the reverse direction with existing pairs

The registered dose ladder tests whether a small share of reversed pairs preserves reverse success. Main-grid evaluations are complete for eligible cells; the final dose analysis must combine reverse recovery, forward retention, seed ranges, and realized budgets.

Where the knee is. We define the knee as the smallest dose whose reverse success reaches 90% of MIX50’s in that cell. A point estimate alone does not establish that adjacent doses differ or that a plateau is equivalent.

What it costs. We report forward retention and general-ability changes alongside reverse recovery. Any budget saving is relative to the forward-only training recipe, not the untouched model. Figure 2 shows the current preservation comparison. The analysis distinguishes paired evaluation uncertainty from variation across training seeds and separates observed runtime from token exposure and additional-CE proxies.

6 Boundary conditions: information and inverse difficulty

The format ladder varies whether an inverse is determined, while algebra orientation and automata probe inverse search difficulty. Several ladder

Model	Task orientations	Cells	Threshold met
gemma3-12b	7	14	8
gemma3-4b	7	21	14
llama31-8b	6	12	7
llama32-3b	11	32	18
olmo2-1b	4	12	5

Table 1: Audited model–task–seed cells. Threshold met means SFT reverse strict success is at most half the same-pass base rate. Counts are descriptive and are not independent task counts or pooled significance tests.

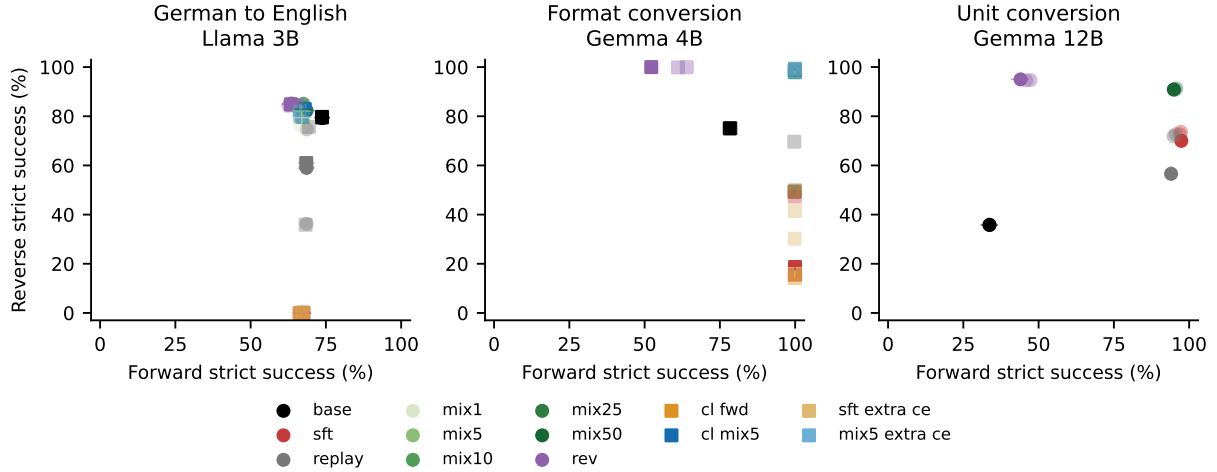


Figure 2: Current preservation frontiers on the fixed reporting panel. Circles show ordinary SFT/dose campaigns; squares show separately generated contrastive/control campaigns. Opaque points and marginal cluster-bootstrap intervals are seed-17 results; translucent points show other training seeds. Contrasts remain paired within each campaign and are retained with measured runtime/exposure records in the source-backed analysis. The units null-collapse boundary case is included. No cross-campaign pooled test is implied.

cells fail eligibility and cannot support tuned-model comparisons; their descriptive base results remain separate.

Two kinds of hardness. Information loss can prevent a unique inverse even with unlimited search, whereas a determined inverse can remain computationally difficult. Failed gates do not distinguish these explanations by themselves.

7 Is the reverse capability still accessible?

Four instruments, reported as separate verdicts rather than averaged: they measure different things, and a disagreement between them is a finding rather than noise.

Elicitation. The audited reports include both prompt recovery and little recovery under tested prompts. Best-prompt summaries are exploratory; behavioral recovery does not prove a specific internal mechanism.

Adapter scaling. Scaling curves retain forward and reverse success together. Forward-progress ratios are undefined when the full update has no positive forward gain, and a curve without reverse loss cannot locate a collapse threshold.

Relearning against a never-had control. Relearning curves are matched to the never-had control by model and seed. A small-sample recovery advantage is descriptive evidence of accessible capability; its uncertainty and starting loss must accompany the comparison. This is also where the contrast with the reversal curse (Berglund et al., 2024) becomes experimental rather than rhetorical: that literature describes capabilities never acquired, and our never-had control is that case, constructed.

Update repair. LoRA spectral sweeps are exploratory median-relative filters on an already low-rank update, with no full-rank noise bulk. Selecting a best threshold on test success is exploratory. Full-FT projection repair uses a separately registered fixed threshold and has produced 1 check-

point passing saved-weight audits. The completed fresh generation campaign improves forward and reverse success relative to the original full-SFT checkpoint. This single exploratory repair does not establish a general remedy.

8 Consequences for attribution

We distinguish exposure to both paired sides from an objective-specific generation benefit.

A gain from an auxiliary objective can reflect its directional information, its additional computation, or the objective itself. Our paired contrastive arms combine forward-only or mixed-direction CE with symmetric representation alignment. Both sides are encoded independently, and negatives are drawn only from training data. The contrastive term has no reverse generation CE, but it still exposes inverse correspondence; it cannot be described as having no inverse supervision.

Matched comparisons include additional-CE controls and shuffled positive pairs. We measure strict reverse generation, forward retention, realized side-token exposure, and compute; embedding retrieval is secondary. Existing and new adapters must be evaluated together in a fresh pass. The contrastive campaign has completed 12 of 27 registered model–task–seed cells at the current snapshot. Table 2 shows completed exploratory contrasts without pooling models or seeds. An interval spanning zero is not an equivalence result. The extra-CE controls match a declared exposure proxy, not measured identical FLOPs or wall time. The existing attribution results require an audit of auxiliary exposure and withdrawn artifacts before supporting a numerical conclusion.

The completed controls qualify the interpretation: mixed-direction alignment sometimes improves reverse generation beyond the extra-CE proxy on format conversion, but the benefit varies across seeds and models. Forward-only alignment provides no consistent advantage over the additional-compute and shuffled-pair controls. For forward-only translation systems at the observed reverse-generation floor, degenerate bootstrap intervals do not establish equivalence. These are interim, within-cell observations; forward retention and the registered extensions are required before a broader objective-specific conclusion.

New-domain eligibility. The 6 original small-model task–model gates failed the unchanged eligibility criteria (Table 4). Complete test-set gates

and scorer audits are retained. No tuned pilot is admitted for these failed cells. Explicit-output variants reuse their corpora; they are prompt diagnostics and cannot be counted as independent new-domain successes. The 6 completed explicit-format gates also fail the unchanged criteria. Larger-model feasibility gates on the original tasks are separately registered. Gemma12B passes the unit-conversion gate and completes valid replicated campaigns (Table 5). Forward-only SFT improves both directions; this is retained as a replicated null-collapse result. Replication seeds are admitted by validity, independently of effect sign. Llama8B also passes Python–C++ translation after a technical parser retry; its pilot is evaluated under the registered same-pass protocol. Boolean gates remain failed, with correctness criteria unchanged.

Full-weight checks. Full-FT training is complete; 6 of 6 validated evaluation campaigns are complete in the current snapshot. Full checkpoints require isolated engines, unlike the single-engine multi-LoRA comparisons. Their newly generated campaigns use the same instances, messages, tokenizer, and scoring settings but span engine restarts. We record this layout and interpret small differences against the decoding floor. Projection-matrix spectral repair is exploratory; it does not assume a validated noise model. Repair results require joint reverse/forward interpretation.

9 Related work

Prior directional degradation and reverse training are established starting points. Our contribution is the joint preservation comparison and controlled attribution, qualified by failed gates and replicated null-collapse results.

Directional degradation and reverse training. Nikiema et al. (2025) study cognitive specialization in code transformation; Zhu et al. (2024) examine direction-dependent degradation in translation. We ask how preservation varies across domains, doses, models, and objectives. Program-execution reasoning also exhibits directional asymmetry (Hasanov et al., 2026), motivating cell-specific measurement of starting reverse capability. The reversal curse concerns capabilities never acquired (Berglund et al., 2024); reverse training addresses this at the data level under data-

Task	Model	Seed	Δ CL-fwd versus SFT	Δ CL-mix5 versus mix5
fmt	gemma3-4b	17	-2.88 [-3.88, -1.96]	+1.08 [+0.32, +1.88]
fmt	gemma3-4b	42	-1.28 [-1.96, -0.64]	-1.04 [-1.80, -0.32]
fmt	gemma3-4b	1234	+2.36 [+1.60, +3.20]	-0.32 [-0.92, +0.24]
fmt	llama32-3b	17	-0.24 [-0.56, +0.04]	+0.00 [-1.12, +1.12]
fmt	llama32-3b	42	-1.48 [-2.12, -0.92]	+1.68 [+0.52, +2.88]
fmt	llama32-3b	1234	-0.52 [-1.00, -0.04]	+4.44 [+3.08, +5.84]
mt_de-en	gemma3-4b	17	+0.00 [+0.00, +0.00]	+0.10 [-2.27, +2.27]
mt_de-en	gemma3-4b	42	+0.00 [+0.00, +0.00]	+0.40 [-2.08, +2.77]
mt_de-en	gemma3-4b	1234	+0.00 [+0.00, +0.00]	-0.30 [-2.47, +2.17]
mt_de-en	llama32-3b	17	+0.00 [+0.00, +0.00]	+0.89 [-1.38, +3.26]
mt_de-en	llama32-3b	42	+0.00 [+0.00, +0.00]	+2.47 [+0.00, +5.04]
mt_de-en	llama32-3b	1234	-0.10 [-0.40, +0.00]	+0.10 [-2.57, +2.87]

Table 2: Completed exploratory reverse-generation contrasts (percentage points), with 99% paired cluster-bootstrap intervals. Five primary contrasts are adjusted within each cell; no adjustment across cells. The original small-model phase is complete; newly registered scale/domain extensions are pending.

Task	Model	Seed	CL-fwd vs extra CE	CL-fwd vs shuffled	CL-mix5 vs extra CE
fmt	Gemma 4B	17	+0.04 [-0.84, +0.96]	-29.20 [-31.80, -26.72]	-0.08 [-0.72, +0.56]
fmt	Gemma 4B	42	-16.04 [-18.00, -14.20]	-19.44 [-21.52, -17.48]	-0.44 [-1.28, +0.44]
fmt	Gemma 4B	1234	+8.20 [+6.80, +9.64]	-22.16 [-25.08, -19.24]	-0.60 [-1.24, +0.00]
fmt	Llama 3B	17	-0.48 [-0.96, -0.08]	-0.68 [-1.16, -0.24]	+1.76 [+0.56, +2.96]
fmt	Llama 3B	42	-1.76 [-2.48, -1.12]	-3.40 [-4.32, -2.52]	+0.64 [-0.44, +1.68]
fmt	Llama 3B	1234	-0.36 [-0.88, +0.12]	+0.28 [+0.08, +0.60]	+2.76 [+1.48, +4.00]
mt_de-en	Gemma 4B	17	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	-0.59 [-2.77, +1.58]
mt_de-en	Gemma 4B	42	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [-2.27, +2.27]
mt_de-en	Gemma 4B	1234	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+1.09 [-1.09, +3.36]
mt_de-en	Llama 3B	17	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+0.69 [-1.68, +3.16]
mt_de-en	Llama 3B	42	+0.00 [+0.00, +0.00]	-0.10 [-0.40, +0.00]	+1.78 [-0.59, +4.25]
mt_de-en	Llama 3B	1234	-0.10 [-0.40, +0.00]	-0.10 [-0.40, +0.00]	-0.69 [-3.26, +1.88]

Table 3: The remaining three registered primary reverse-generation contrasts, in percentage points with 99% paired cluster-bootstrap intervals. Together with Table 2, these display all five primary comparisons for every completed cell. Extra CE is an exposure/compute proxy, not matched measured FLOPs. Degenerate zero intervals at an observed generation floor do not establish equivalence. Results remain exploratory; new scale/domain extensions are pending.

and compute-matched comparisons (Golovneva et al., 2024). Our relearning experiment compares a collapsed model with a model-matched never-had control. Replacement mixtures hold instances and steps fixed while auditing realized exposure.

Forgetting and rehearsal. Broader forgetting after fine-tuning (Luo et al., 2023) and differences between LoRA and full fine-tuning (Biderman et al., 2024) motivate our general-ability and full-weight controls; they do not bound directional loss in these tasks. Rehearsal can preserve earlier performance (Scialom et al., 2022). Generic replay at the registered replacement share tests whether directional content adds preservation beyond that explanation. Concentration of inverse loss relative to general ability remains a separate empirical question.

Recovery and task inference. Wrapper and off-set accounts of fine-tuning (Jain et al., 2024; Lee

et al., 2024), update filtering (spe, 2026), and skewed task inference (Kotha et al., 2024) motivate distinct recovery instruments. Our directional tests include a never-had control, both-direction retention, and fixed-threshold exploratory repair. Behavioral recovery does not identify a unique internal mechanism, and bidirectional behavior need not imply a unified representation (lat, 2026).

Attributing objective gains. Data mixtures and reporting choices can confound objective attribution (Longpre et al., 2023; Lipton and Steinhardt, 2018). RevThink adds backward questions and reasoning (Chen et al., 2025); its cross-entropy components illustrate why directional exposure must be controlled when assessing a bidirectional objective. Our paired contrastive comparison includes reversed-pair SFT, extra-CE proxies, and shuffled positives, measuring generation in both directions. These controls test the scope of an

Task	Model	Forward	Reverse	Eligibility
units	llama32-3b	0.4	1.9	Fail
units	gemma3-4b	37.6	3.5	Fail
logic	llama32-3b	0.0	0.0	Fail
logic	gemma3-4b	0.0	0.0	Fail
py_cpp	llama32-3b	78.6	74.1	Fail
py_cpp	gemma3-4b	5.1	99.9	Fail

Table 4: Original new-domain base gates: strict success in percent. Failed cells are retained and receive no tuned pilot. Explicit-output diagnostics use the same semantic splits and are separate exploratory variants.

Task	Model	Seed	Arm	Forward	Reverse
py_cpp	llama31-8b	17	base	85.9	100.0
py_cpp	llama31-8b	17	sft	100.0	0.0
py_cpp	llama31-8b	17	mix50	100.0	48.7
py_cpp	llama31-8b	17	replay	100.0	97.6
py_cpp	llama31-8b	17	rev	0.0	0.0
units	gemma3-12b	1234	base	33.7	35.8
units	gemma3-12b	1234	sft	96.9	72.7
units	gemma3-12b	1234	mix50	95.1	90.7
units	gemma3-12b	1234	replay	94.8	71.9
units	gemma3-12b	1234	rev	47.3	94.7
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	mix50	94.9	90.9
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	rev	44.0	95.0
units	gemma3-12b	42	base	33.7	35.8
units	gemma3-12b	42	sft	97.3	73.7
units	gemma3-12b	42	mix50	95.7	91.3
units	gemma3-12b	42	replay	95.5	72.7
units	gemma3-12b	42	rev	45.9	94.6

Table 5: Eligible larger-model generality campaigns: strict success in percent, with all arms and the base evaluated in one fresh pass per model/task/seed. The completed unit-conversion replications improve both directions under SFT and are retained null-collapse boundary cases. Additional eligible campaigns enter this table after validated analysis.

objective-specific gain without assuming that auxiliary objectives are universally unnecessary.

10 Conclusion

Paired training data makes direction a variable that can be changed without collecting new pairs. Evaluating only the trained direction leaves its inverse unchecked. Our study is organized around preservation, boundary conditions, and objective attribution: how much can ordinary reversed-pair SFT accomplish, and what do more elaborate methods contribute beyond that baseline? The final conclusions will follow the audited generation results, including null effects and failed eligibility gates.

Limitations

Where the repair does not reach. Reversed-pair training cannot recover information absent from an input. Our failed new-domain gates also limit generality beyond established tasks.

Scale. The full arm set runs at 1–4B with three seeds; only core arms run at 8–12B with two where eligible in the original grid. Separately registered larger-model extensions add replication and objective controls in selected cells; coverage remains uneven. This does not establish scaling to larger models.

LoRA. Most arms use LoRA. Prior work reports different retention under full fine-tuning (Biderman et al., 2024); this does not guarantee a conservative directional-loss estimate in our tasks. The full-FT check covers only the registered Llama 3B seed-17 cells; it cannot bound directional loss across other models or training recipes.

Resolution of the dose ladder. Adjacent rungs differ by a couple of points, which our evaluation sets may not separate reliably. Final dose claims require intervals and seed ranges; an interval spanning zero does not establish equivalence.

Task	Seed	Direction	Base	Full SFT	Full mix5
code	17	forward	37.9	70.0	70.7
code	17	reverse	31.9	2.0	65.9
mt_de-en	17	forward	73.3	70.9	72.0
mt_de-en	17	reverse	80.1	72.8	83.6
mt_en-de	17	forward	80.1	87.0	86.6
mt_en-de	17	reverse	72.9	69.9	72.4
mt_en-zh	17	forward	73.9	79.1	79.2
mt_en-zh	17	reverse	75.9	66.9	71.8
mt_zh-en	17	forward	76.3	74.2	73.9
mt_zh-en	17	reverse	73.8	70.2	78.5
sql	17	forward	49.3	60.5	59.7
sql	17	reverse	50.4	18.7	55.1

Table 6: Completed Llama3B full-weight campaign rates in percent, with forward and reverse retention. Checkpoints use isolated engines and identical held-out instances; these comparisons span engine restarts. Paired intervals are recorded in followup contrasts. An empty body means no validated full campaign has completed.

Criteria with a model in them. Two directions are scored by round-tripping through a frozen model rather than by execution. Each reports that model’s own accuracy on the references as the criterion’s ceiling where measured. Extractor/parser errors can reject valid generations; the D2T measurement is still undergoing validation.

Languages and domains. The original code-transformation cell is Python only; the translation cells cover two language pairs, both with English on one side. Synthetic task variants do not establish natural code or unrestricted reasoning generality, and reused semantic corpora are not independent evidence. Independent real-data transfer and alternate-prompt/training-recipe tests are drafted in Appendices A.3 and A.4; they have no measured results yet.

References

2026. [The illusion of latent generalization: Birectionality and the reversal curse](#). *Preprint*, arXiv:2604.04943. Probes are consistent with forward and reverse stored as distinct entries rather than one direction-agnostic representation.
2026. [Spectral unforgetting: Post-hoc recovery of damaged capabilities without retraining](#). *Preprint*, arXiv:2605.20296.
- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. 2024. [The reversal curse: Llms trained on “a is b” fail to learn “b is a”](#). In *International Conference on Learning Representations*.
- Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, et al. 2024. Lora learns less and forgets less. *Transactions on Machine Learning Research*.

Read 2026-09-11. LoRA underperforms full fine-tuning on target domains but better maintains out-of-domain performance. Qualifies every LoRA-based claim here.

Justin Chih-Yao Chen, Zifeng Wang, Hamid Palangi, Rujun Han, Sayna Ebrahimi, Long T. Le, Vincent Perot, Swaroop Mishra, Mohit Bansal, Chen-Yu Lee, and Tomas Pfister. 2025. [Reverse thinking makes llms stronger reasoners](#). In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*. Measures FORWARD accuracy: backward data added to improve forward reasoning. Loss is a sum of three cross-entropy terms over a union of formats (p. 4), and no baseline holds data direction fixed.

Olga Golovneva, Zeyuan Allen-Zhu, Jason Weston, and Sainbayar Sukhbaatar. 2024. [Reverse training to nurse the reversal curse](#). *Preprint*, arXiv:2403.13799.

Eshgin Hasanov, Md. Mahadi Hassan Sibat, Shubhra Kanti Karmaker, and Aashish Yadavally. 2026. [The path not taken: Duality in reasoning about program execution](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. DexBench: 445 paired instances, 13 LLMs evaluated. No fine-tuning. Code and data at <https://github.com/sail-ucf/dexbench>.

Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, et al. 2024. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. In *International Conference on Learning Representations*. Read 2026-09-11. Fine-tuning rarely alters underlying capabilities; a minimal ‘wrapper’ is learned on top.

Suhas Kotha, Jacob Mitchell Springer, and Aditi Raghunathan. 2024. Understanding catastrophic forgetting in language models via implicit inference. In *International Conference on Learning Representations*. Read 2026-09-11. Fine-tuning skews IMPLICIT TASK INFERENCE; conjugate prompting

recovers pretraining capability. A precedent for H1, not merely for forgetting.

Andrew Lee, Xiaoyan Bai, Itamar Pres, et al. 2024. A mechanistic understanding of alignment algorithms: A case study on dpo and toxicity. In *International Conference on Machine Learning*. Read 2026-09-11. Pretraining capabilities are NOT removed but BYPASSED, via an offset distributed amongst layers. Bears on the layer-ablation experiment.

Zachary C. Lipton and Jacob Steinhardt. 2018. Troubling trends in machine learning scholarship. *arXiv preprint arXiv:1807.03341*. Read 2026-09-11. Names ‘failure to identify the sources of empirical gains’ as a trend; our attribution section is an instance with direction in place of hyper-parameters.

Shayne Longpre, Le Hou, Tu Vu, et al. 2023. The flan collection: Designing data and methods for effective instruction tuning. In *International Conference on Machine Learning*. Read 2026-09-11. Task balancing and enrichment are critical and overlooked; gains attributed to method often belong to the mixture.

Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2023. [An empirical study of catastrophic forgetting in large language models during continual fine-tuning](#). *Preprint*, arXiv:2308.08747. Read 2026-09-11. Forgetting across 1b-7b during continual instruction tuning, worsening with scale in that range.

Serge Lionel Nikiema, Jordan Samhi, Micheline Moumoula, Albarick Djié, Abdoul Kader Kaboré, Jacques Klein, and Tegawendé F. Bissyandé. 2025. [Using contrastive learning to improve two-way reasoning in large language models: The obfuscation task as a case study](#). Names “cognitive specialization”: forward-only fine-tuning degrades pre-existing bidirectional reasoning. Declares a Bidirectional Fine-Tuning baseline (p. 6) and reports no number for it.

Thomas Scialom, Tuhin Chakrabarty, and Smaranda Muresan. 2022. Fine-tuned language models are continual learners. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Read 2026-09-11. 1 percent memory buffer preserves 100 percent of prior performance across 8 sequential tasks. The buffer replays ACTUAL earlier-task examples and ADDS to the budget; our dose is the same pairs reversed and REPLACES. Our replay arm is their rehearsal, and the prediction is that it does NOT cure directional collapse.

Dawei Zhu, Pinzhen Chen, Miaoran Zhang, Barry Haddow, Xiaoyu Shen, and Dietrich Klakow. 2024. [Fine-tuning large language models to translate: Will a touch of noisy data in misaligned languages suffice?](#) In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Single-direction fine-tuning generalizes, EXCEPT

that English on the target side causes task misinterpretation damaging translation into non-English.

A Paper-completion protocol and reporting plan

This appendix records frozen exploratory additions registered before their measurements. Existing new-domain and contrastive campaigns retain priority. Allocation-local smokes precede automatic admission of each new production workload. Registration and queue admission do not establish completed results. Visible red markers reserve conclusions to write after analysis. Failed gates and the replicated Gemma12B unit-conversion null-collapse result remain visible.

A.1 Dose, forward retention, and computational cost

The main figure plots forward and reverse strict success for the base model, forward SFT, generic replay, available reversed-pair doses, and contrastive/control arms. Panels retain model, task orientation, and seed identity. Arms lacking a common evaluation campaign are not connected as paired observations. The units boundary case is shown alongside collapse cells. Repeated passes use the latest validated completion time, with earlier passes retained in the audit; selection never uses effect sign.

Paired contrasts use the domain’s instance/cluster unit. Training-seed variation is separate: resampling test instances does not measure training variability. No aggregate pools cells as independent domains. The dose knee uses the criterion in §5; overlapping intervals do not establish equivalence.

The cost table reports available trainer runtime, optimizer steps, and CE-corpus tokens. Auxiliary encoded-side exposure is retained separately in the analysis. Missing timing, GPU-type, queue-time, or exposure records remain unavailable. Additional CE is the declared proxy; corpus token counts alone do not establish matched objective FLOPs or wall time.

Result to fill. `<<dose-sentence>>`

A.2 General ability and overlap controls

The fixed panel is German-to-English/Llama3B, format conversion/Gemma4B, and units/Gemma12B. Each campaign compares base, SFT, replay, mixed-direction SFT, forward-only and mixed-direction contrastive arms, and

Registered evaluation	Model/task cells	Training seeds	Campaigns
General probes	3	17,42,1234	9
Prompt/template tests	4	17,42,1234	12
Independent transfer	2	17,42,1234	6
Recipe evaluations	2	17,42,1234	6

Table 7: Frozen paper-completion panel. Probe corpora retain 541 IFEval prompts and 1319 GSM8K questions; independent transfer retains 1688 OPUS translation pairs. Synthetic magnitude/template tests retain 1000 instances each. Recipe packs train 48 new adapters in isolated directories. Counts are planned workloads and audited corpus sizes, not completed results.

Task	Model	Seed	Arm	Forward (%)	Reverse (%)
mt_de-en	llama32-3b	17	base	73.7	79.2
mt_de-en	llama32-3b	17	sft	67.1	0.0
mt_de-en	llama32-3b	17	replay	68.6	59.0
mt_de-en	llama32-3b	17	mix5	67.4	82.0
mt_de-en	llama32-3b	17	mix50	67.4	82.9
units	gemma3-12b	17	base	33.7	35.8
units	gemma3-12b	17	sft	97.4	70.0
units	gemma3-12b	17	replay	94.0	56.6
units	gemma3-12b	17	mix50	94.9	90.9

Table 8: Representative seed-17 preservation rates. Each model/task row set comes from one complete campaign. Additional doses, seed variation, paired intervals and measured cost records are retained in the source-backed analysis. The units boundary case is included independently of effect sign.

their additional-CE controls. The auxiliary arm is fixed rather than selected by its best test score.

IFEval measures instruction following without demonstrations; GSM8K measures numerical reasoning using fixed training demonstrations. A neutral assistant system prompt is shared by all systems. Dataset revisions and overlap exclusions are recorded before generation. Auditing covers both sides of every local training/validation corpus, replay, and mixed-task sources, using normalized exact/contained strings. Fixed demonstrations are checked against the retained test set. Lexical auditing does not establish semantic or pretraining independence; shared numerical structure limits interpretation of the units/GSM8K comparison.

All systems are evaluated together in fresh campaigns with adapter-effectiveness checks. Report strict IFEval and strict GSM8K endpoints separately; loose instruction checks and flexible numeric extraction remain secondary. Pair endpoint changes within each campaign and show seed variation. A concentration claim is scoped to this declared panel; unrelated benchmark scales are not subtracted as if interchangeable.

A.3 Independent real-data transfer

Existing German-to-English checkpoints are evaluated on the independently sourced OPUS-100 German-English test split.¹ The panel includes Llama3B and Gemma4B with the registered seeds and the same SFT, replay, mixed-direction, contrastive, and additional-CE systems. Dataset revision, character-length inclusion rules, directions, overlap exclusions, and scorer are frozen before generations. A new prompt on an existing semantic corpus remains a diagnostic rather than independent transfer evidence.

Overlap auditing includes all local training, validation, replay, and original test sources. Gold, echo, empty, and invalid-output oracles run on compute. Original model-specific COMET thresholds are retained. Strict success, continuous COMET, chrF++, echo, and off-target rates remain separate; checkpoint choice, decoding, and scoring are not tuned on this corpus.

Generate and score the full fresh base first, using the resident engine retained for tuned systems. Failure of the unchanged eligibility conjunction remains boundary evidence and blocks tuned generation. Small smokes establish engineering behavior, not production eligibility. Report paired for-

¹<https://huggingface.co/datasets/Helsinki-NLP/opus-100>.

Result to fill. `<<probe-sentence>>`

Task	Model	Arm	Minutes	Steps	CE corpus tokens
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	mix50	9.0	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	sft	8.9	306	1096262
mt_de-en	llama32-3b	mix5	8.9	306	1096262
mt_de-en	llama32-3b	replay	8.9	306	1096110
mt_de-en	llama32-3b	cl_fwd	241.9	306	1096262
mt_de-en	llama32-3b	cl_mix5	62.2	306	1096262
mt_de-en	llama32-3b	sft_extra_ce	123.2	306	1096262
mt_de-en	llama32-3b	mix5_extra_ce	119.1	306	1096262
fnt	gemma3-4b	sft	30.6	306	1927990
fnt	gemma3-4b	mix5	30.7	306	1927882
fnt	gemma3-4b	replay	30.2	306	1927978
fnt	gemma3-4b	cl_fwd	194.3	306	1927990
fnt	gemma3-4b	cl_mix5	193.5	306	1927882
fnt	gemma3-4b	sft_extra_ce	191.3	306	1927990
fnt	gemma3-4b	mix5_extra_ce	194.0	306	1927882
units	gemma3-12b	sft	21.6	306	788819
units	gemma3-12b	mix50	22.1	306	788819
units	gemma3-12b	replay	21.5	306	799547

Table 9: Recorded seed-17 trainer runtimes (minutes), optimizer steps, and sequence tokens in the tokenized CE corpus (before epoch repetition). Auxiliary encoded-side exposure remains separately recorded in the analysis; these token counts are not total objective FLOPs. Runtime varies with GPU/training layout and is not a claim of matched wall time. Missing measurements are unavailable, never inferred.

ward/reverse changes only for eligible campaigns.

Result to fill. ⟨⟨transfer-sentence⟩⟩

A.4 Prompt and training-recipe robustness

The prompt panel includes translation/Llama3B, format conversion/Gemma4B, units/Gemma12B, and Python-C++/Llama8B. Frozen variants retain the original primary instruction and add fixed task/contract prefixes. Both directions use the same semantic instances, decoding limits, and scorers. Keep every frozen prompt, separating echo/off-target output, semantic error, and measurement failure. Best-prompt summaries remain exploratory recovery diagnostics.

Synthetic extrapolation additionally uses exact unit magnitudes outside original support and behavior-disjoint piecewise-quadratic Python/C++ functions. Original allowlisted scorers and input ranges remain unchanged; gold outputs are audited on all generated instances on compute. These tests do not establish unrestricted natural-code, loop/list, or independent-domain generality.

The recipe panel fixes lower/higher learning-rate and LoRA-rank variants around the original recipe, varying the factors separately. Parameter values, cells, seeds, and primary contrasts are fixed in the recorded protocol. Preserve alpha/rank ratio, effective batch, paired-data identity, direc-

tion mixture, sequence limit, epoch count, and optimizer schedule. Audit realized exposure, step count, and GPU cost. SFT and mixed-direction SFT are compared on format conversion/Llama3B and units/Gemma12B, retaining collapse and null cases.

Store new adapters separately. Regenerate the original arms, base, and all recipe variants together in each campaign. Retain every adverse/null outcome and report paired intervals and training-seed ranges; no best-recipe selection supports the primary claim.

Result to fill. ⟨⟨robust-sentence⟩⟩

A.5 Contribution and reproducibility

The final abstract and conclusion should state the supported contribution relative to prior directional degradation and reverse-training work: the preservation frontier, direction and compute controls for objective attribution, and limits revealed by gates and null cells. Generality, disproportionate loss, inexpensive preservation, and recovery are separate claims with separate evidence requirements. A favorable contrastive result supports a scoped objective contribution; a null or generation-floor result does not establish equivalence.

Final contribution wording to fill. ⟨⟨claim-sentence⟩⟩

Deliverable	Required evidence	Current status
Preservation figure	Both directions, intervals, seeds, doses, exposure and measured cost	Audited synthesis integrated; interpretation pending
General-ability table	Frozen panel, overlap exclusions, same-pass changes and intervals	Registered and smoke-gated; measurements pending
Independent transfer	Corpus/version, eligibility, both directions, failures and hashes	Registered OPUS protocol; measurements pending
Robustness table	Frozen prompts/recipes, both directions, costs and all outcomes	Registered panel; measurements pending
Reproducibility manifest	Splits/scorers, environment, manifests, trial hashes and analysis commands	Local provenance retained; release audit pending

Table 10: Reporting checklist. Registration, auditing, and queue admission are separate from completed experimental evidence.

The local reproducibility package includes permitted corpora or versioned reconstruction recipes; overlap and scorer audits; manifests and environment settings; irreproducible per-instance generations and their hashes; analysis entryptpoints; and figure/table provenance. Failed gates, withdrawals, and corrected replacements remain traceable. Dataset/weight redistribution follows their licenses.

Before release, replay analysis and the PDF build from a clean environment, check reported values against generating trials and references, and remove identifying local names/paths while retaining a hash mapping for transformed artifacts. Missing historical code versions must be recorded rather than implying exact training reproducibility. This is a local artifact deliverable; no release or submission has been performed.